

They're all powered by the same technology that made the digital revolution possible by making all devices portable and compact in a cost effective manner – the microcontroller revolution. And that technology is the microprocessor.

Microprocessors lie at the heart of every device that enables us to be our virtual/digital selves. In addition, general and specialised microprocessors are used to create a large variety of other products. It is due to the microcontroller that 3D printing has become available to the common man. They're undoubtedly the first choice for student projects and budding entrepreneurs to develop models faster than previously imaginable.

A 'microcontroller' is yet another version of a processor that has enabled the production of a wide variety of embedded devices such as the printer.



A microcontroller

With increased capabilities and increasing simplicity, the use of microcontroller is shifting from the experts to the common man. This has enabled the likes of the tinkerer next door to go one step ahead of the usual printer and build a 3D printer just as a hobby.

To put it simply, a microcontroller is a “processor that concentrates on one

single task and allows other devices to connect for direct control of hardware (such as a robot) or via peripheral devices”. In contrast, if hardware control is via a computer at home, we need to connect the computer to the microcontroller and then instruct the microcontroller to control the hardware.

The topic of microcontrollers is taught in the second or third year of Engineering courses in India. But its increasing simplicity has enabled even schools to teach the subject to students using microcontrollers for their projects. This helps give the students a glimpse into how things around us are built by tech developers.

They're everywhere

The capabilities of electronic devices surrounding us have increased by leaps and bounds. Mobile phones including smartphones, printers and tablets